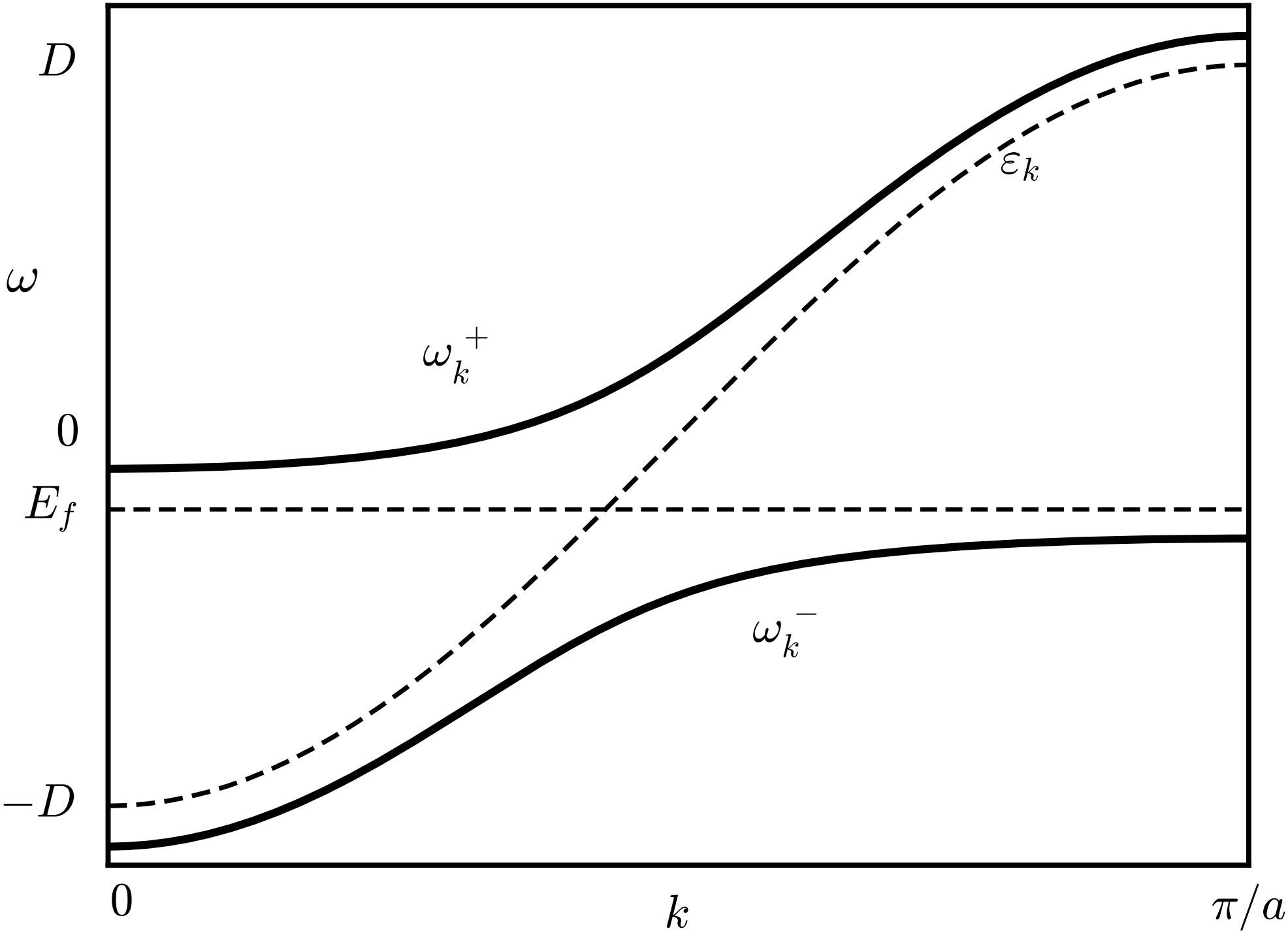
D
ω
0
E_f
$-D$
ε_k
ω_k^+
ω_k^-
0
k
π/a